ELEC226/326 001 – PROBABILITY & RANDOM PROCESSES

Course Syllabus – Fall 2026

TEACHING TEAM 2

Land Acknowledgement 2
Inclusivity Statement 2
Course Instructor 2
Graduate Teaching Assistants 2

ELEC226/326 (F 3-0-0.5 3.5) 3

Course Description 3
Mode of Instruction 3
Course Location and Time 3
Pre-Requisite Knowledge 3
Course Learning Outcomes (CLO) 3
Course Evaluation 4
Assessment Descriptions 5
Grading 5
Course Materials 5
Weekly Course Learning Outcomes 7

COURSE COMMUNICATION 9

Questions About Course Material 9
Course Announcements 9
Office Hours 10
Confidential Matters 10
Absences (Academic Considerations) and missed assignments 10
Late Submission Policy 10
Requesting Academic Consideration Related to Group Work 10

STANDARD QUEEN'S AND SMITH ENGINEERING POLICIES 10

Netiquette 10
Student Code of Conduct 11
Copyright 11
Academic Integrity 11
Invalid Exams 12
Absences - Academic Considerations 13
Absences – Requests for Excused Absences for Significant Events (REASE) 13
Academic Accommodations 13
Religious Accommodations 13
Other human-rights based Accommodations 14
Accommodations for Disabilities 14
Student Wellbeing 15
Educational Technology 15
Technical Support 15

This is your course syllabus. Please download the file and keep it for future reference.

TEACHING TEAM

LAND ACKNOWLEDGEMENT

Queen's University is situated on traditional Anishinaabe and Haudenosaunee Territory. See: http://www.queensu.ca/encyclopedia/t/traditional-territories

INCLUSIVITY STATEMENT

Queen's students, faculty, and staff come from every imaginable background – small towns and suburbs, urban high rises, Indigenous communities, and from more than 100 countries around the world. You belong here: https://www.queensu.ca/vpcei/support/overview.

COURSE INSTRUCTOR

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GRADUATE TEACHING ASSISTANTS

- Rana Gharib
- Hamza Saleem
- Mehran Rasheed

ELEC226/326 (F 3-0-0.5 3.5)

Course Description

This course introduces probabilistic models and methods for addressing uncertainty and variability in engineering applications. Topics include sample spaces and events, axioms of probability, conditional probability, independence, discrete and continuous random variables, probability density and cumulative distribution functions, functions of random variables, and random processes.

(31/0/0/11/0) (Mathematics/Natural Sciences/Complementary Studies/Engineering Science/Engineering Design)

Mode of Instruction

The delivery mode of this course is ***in-person***.

Course Location and Time

Lectures			
ELEC226 & ELEC326 001	**Days & Times**	**Room**	**Class Dates**
	Mo 10:30AM - 11:30AM	WALTER LIGHT RM205	Sep 8, 2026-Dec 8, 2026
	We 9:30AM - 10:30AM	WALTER LIGHT RM205	Sep 8, 2026-Dec 8, 2026
	Fr 8:30AM - 9:30AM	WALTER LIGHT RM205	Sep 8, 2026-Dec 8, 2026

Tutorials			
ELEC226	Fr 10:30AM - 11:30AM	BOTTER RM B139	Sep 8, 2026-Dec 8, 2026
ELEC326 001	We 11:30AM - 12:30PM	WALTER LIGHT RM210	Sep 8, 2026-Dec 8, 2026

Pre-Requisite Knowledge

This course is designed for learners with some background on Calculus.

Prerequisite courses: **APSC171**

Course Learning Outcomes (CLO)

By the end of this course, students should be able to:

CLO	Description	Indicator[1]
CLO 1	Describe different types of random variables and identify import Distributions.	1
CLO 2	Discuss the law of large numbers and asymptotic behaviors, and the concept of information and entropy.	1 & 4
CLO 3	Characterize and identify distribution of functions of multiple random variables and well-known random and point processes.	1
CLO 4	Apply probability theory for modeling uncertainties involved in engineering problems.	3
CLO 5	Solve problems involving probability with conditional probabilities.	2 & 3
CLO 6	Calculate different statistics of random variables by manipulating one and two random variables and working with joint distributions	3

COURSE EVALUATION

Assessment Weighting

Assessment Tool	Date	Weight	Alignment with CLOs
In-Class Tests	**Weeks 4, 7, 10**	**45%**	**1,2,3,4,5,6**
Test 1	Friday of Week 4 (Oct. 02)	15%	1,3,4,5,6
Test 2	Friday of Week 7 (Oct. 30)	15%	1,2,3,4,5,6
Test 3	Friday of Week 10 (Nov. 20)	15%	1,2,3,4,5,6
Final Exam	**Please see the Final Exam schedule posted by OUR Exams Office**	**55%**	**1,2,3,4,5,6**
		100%	

[1] For further information on graduate attributes and indicators, please click here.

Assessment Descriptions

Tests

There are three closed-book, in-class tests in this course. These tests are designed to provide students with timely feedback on their understanding of the course material. All three tests will count toward the calculation of your final course grade.

Assignments

There are weekly assignments (not for submission) with solutions which can be found on OnQ.

Final Exam

The final exam is closed book. Students must write their exam on the day, and time scheduled by the University. You should not schedule vacations, travel, etc. during the exam period. The Term and Session Dates will indicate the final exam period session dates in each term.

Grading

All assessments in this course will receive numerical percentage marks. The final grade you receive for the course will be derived by converting your numerical course average to a letter grade according to the established Grade Point Index.

Feedback on Assessments

The teaching team will provide feedback on graded activities. You can expect feedback on your assessments within seven days of the due date.

Accessing Your Final Grade

Your final grades will show on SOLUS. Official transcripts showing final grades will be available on the Official Grade Release Date. Please note that in official transcripts, a mark of IN (incomplete) is considered a grade, and your transcript is released with this grade.

Course Materials

Required Textbook

There is no required textbook for this course. Lecture notes are posted on OnQ.

Optional Textbook

Probability and Stochastic Processes: A Friendly Introduction for Electrical and Computer Engineers (3rd Edition), Roy D. Yates and David J. Goodman.

Cost: $220.95 in paperback and available on Wiley.

Other Material

All other course material is accessible via OnQ.

Required Calculator

A Casio 991 is required. **ONLY** this type of non-programmable, non-communicating calculator will be allowed during tests and exams.

Suggested Time Commitment

This course represents a study period of one term spanning 12 weeks. Learners can expect to invest on average 7-9 hours per week in this course. Learners who adhere to a predetermined study schedule are more likely to successfully complete the course.

WEEKLY COURSE LEARNING OUTCOMES

Week	Learning Outcomes
1	**Introduction to Elements of Probability Theory and Applications for Engineers** • Describe the differences between deterministic and random phenomena [CLO1] • Define the different concepts of a probability model (e.g. event, sample space) [CLO1] • Cite the applications of probability in electrical and computer engineering [CLO4] • Identify the differences between countable and uncountable sets [CLO1] • Use various operations (e.g. union, intersection, etc.) between sets [CLO1]
2	**Probability and Conditional Probability** • Use probability axioms to solve problems [CLO1,5] • Solve problems related to sampling with and without replacement [CLO1,4] • Describe the differences between countable and uncountable sample spaces [CLO1] • Solve problems related to conditional probability [CLO5] • Prove the Bayes' and total probability theorems [CLO1,3,4,5,6]
3	**Discrete and Continuous Random Variables** • Describe the concept of a random variable [CLO1] • Provide examples of a random variable [CLO1] • Define the probability mass function and cumulative distribution function of random variables [CLO1,4,5] • Use the properties of probability mass function and cumulative distribution function of random variables to solve problems [CLO1,4,5] • Describe important discrete random variables (e.g., Bernoulli, degenerate, etc.) [CLO1] • Plot the probability mass functions and cumulative distribution functions of important discrete random variables [CLO1] • Describe events involving various types of random variables [CLO3,4]
4	**Discrete and Continuous Random Variables (continued)** • Describe the important continuous random variables (e.g. uniform, exponential, gamma, Gaussian, etc.) [CLO1] • Plot the probability mass functions and cumulative distribution functions of important variables [CLO1,4] • Calculate the probability of various events using the probability density function and probability mass function [CLO1,4]

Week	Learning Outcomes
5	**Expected Value, Variance and Other Moments of Random Variables** • Calculate the expectation of continuous and discrete random variables using the probability density function [CLO1,2,5] • Discuss the concept of expectation [CLO1,2] • Use the relationship between the probability density function and cumulative distribution function [CLO1,5] • Apply a Chi-square Test [CLO4] • Calculate the variance of different random variables [CLO4] • Determine probability bounds [CLO4] • Compare probability bounds using various inequality theorems in relation to different moments of a random variable [CLO4,6]
6	**Functions of a Random Variable, Moments, Entropy, and Information** • Use different applicable random variable distributions (e.g., uniform, normal, t-distribution, Cauchy distribution, exponential, Gamma, etc.) [CLO1,5] • Apply useful mathematical inequalities (e.g., Markov and Chebyshev, Chi-square Test, Poisson process, etc.) [CLO3,4,5] • Solve problems related to different distributions of random variables [CLO1,3,4, 5]
7	**Functions of a Random Variable, Moments, Entropy, and Information (continued)** • Differentiate between continuous type and discrete type random variables [CLO1] • Solve problems involving conversion of continuous type to discrete type random variables, continuous type to continuous type, and monotone functions of random variables [CLO1,3,4,5] • Identify moments and characteristic functions, entropy, and information [CLO2] • Solve problems related to moments, characteristic functions, entropy, and information [CLO2]
8	**Multiple Random Variable and Decision-Making** • Describe the elements of decision making (e.g., False Alarm, missed detection, Maximum-Likelihood (ML), etc.) [CLO4,5,6] • Use the concept of Multiple Random Variables, joint Probability Distribution Function (PDF) and Cumulative Distribution Function (CDF) [CLO4,5,6]
9	**Pairs of Discrete Random Variables** • Differentiate between marginal Probability Mass Functions (PMFS), Conditional PMFS, jointly continuous random variables, and normal random variables [CLO4,5,6] • Apply the concepts of independence and conditional pdfs [CLO4,5,6]

Week	Learning Outcomes
10	**One Function of Two Random Variables** • Apply the main operations between two random variables such as difference, sums, and multiplication [CLO3,5,6] • Define circularly symmetric, distance, max, and min [CLO3,5,6]
11	**Two Functions of Two Random Variables** • Outline some examples of two functions of two random variables [CLO6] • Find the distribution of the two functions of two random variables using the Jacobian method [CLO6] • Solve problems involving Gaussian random vectors [CLO6] • Transform a pair of two random variables by determining the distribution of multiple functions of multiple random variables [CLO6]
12	**Limit Theorems and Basic Elements of Stochastic Processes** • Demonstrate the expected values of a function of two random variables (discrete and continuous) [CLO6] • Apply the concepts of random processes [CLO3,6] • Apply Central Limit Theorem [CLO6]

COURSE COMMUNICATION

QUESTIONS ABOUT COURSE MATERIAL

Questions or comments regarding the course material that can be of benefit to other students should be posted in the Q&A forum on the class website. The instructor, TAs, and students are encouraged to answer these questions directly in the discussion forum for the benefit of everyone in the course.

COURSE ANNOUNCEMENTS

The instructor will routinely post course news in the Announcements section on the main course homepage on OnQ. Please sign up to be automatically notified by email when the instructor posts new information in the Announcements section. Instructions on how to modify your notifications are found in the **Begin Here** section of the onQ course site.

Office Hours

In addition to interaction in the Q&A discussion forums, you will have the opportunity to interact with either a TA or the instructor during office hours. The instructor will provide a schedule of availability at the beginning of the term.

Confidential Matters

If you have a confidential matter you would like to discuss with your instructor, their contact details are on the first page of this document. Expect email replies within 48 hours.

Absences (Academic Considerations) and Missed Assignments

Students who miss assignments, quizzes, classes, labs, tutorials or other course deliverables due to unforeseen personal circumstances outside of their control can apply for Academic Consideration. For information on academic considerations due to extenuating circumstances, please review the information on the Smith Engineering website. Note that unacceptable reasons include extracurricular activities, work, travel plans, scheduling conflicts, generally being behind on schoolwork, etc. Do not schedule travel during midterms and final exams, as travel and work are not acceptable reasons for granting academic considerations.

Missed in-class test will not be re-taken. For a missed in-class test with appropriate notification, the 15% for the missed test will be added to the weight of the final exam.

Late Submission Policy

In the event of extenuating circumstances, you must follow the policies for requesting an academic consideration (as described above) or engage the appropriate Academic Accommodation process (see Academic Accommodation section below). In the absence of an approved consideration request, or accommodation in VENTUS, the normal course policies for missed assessments will apply.

Requesting Academic Consideration Related to Group Work

This course does not include any group work as part of the assessment.

STANDARD QUEEN'S AND SMITH ENGINEERING POLICIES

Netiquette

In this course, you may be expected to communicate with your peers and the teaching team through electronic communication. You are expected to use the utmost respect in your dealings with your colleagues or when participating in activities, discussions, and online communication.

The following is a list of netiquette guidelines. Please read them carefully and use them to guide your online communication in this course and beyond.

1. Make a personal commitment to learn about, understand, and support your peers.
2. Assume the best of others and expect the best of them.
3. Acknowledge the impact of oppression on the lives of other people and make sure your writing is respectful and inclusive.
4. Recognize and value the experiences, abilities, and knowledge each person brings.
5. Pay close attention to what your peers write before you respond. Think through and re-read your writings before you post or send them to others.
6. It's alright to disagree with ideas, but do not make personal attacks.
7. Be open to be challenged or confronted on your ideas and challenge others with the intent of facilitating growth. Do not demean or embarrass others.
8. Encourage others to develop and share their ideas.

Student Code of Conduct

Queen's University values maintaining an environment free of, and will not tolerate, harassment, discrimination, and reprisal. The Student Code of Conduct applies to all students at Queen's. It outlines the activities and behaviours that could be considered Non-Academic Misconduct (NAM). The Code also describes the NAM process and the sanctions that could be imposed on a student found responsible for a violation. All students should be familiar with the Student code of conduct and related policies on sexual violence prevention and response and harassment and discrimination prevention and response.
https://www.queensu.ca/nonacademicmisconduct/policies

Copyright

Course materials created by the course instructor, including all slides, presentations, synchronous and asynchronous course recordings, handouts, tests, exams, and other similar course materials, are the intellectual property of the instructor. It is a departure from academic integrity to distribute, publicly post, sell or otherwise disseminate an instructor's course materials or to provide an instructor's course materials to anyone else for distribution, posting, sale or other means of dissemination, without the instructor's ***express consent***. A student who engages in such conduct may be subject to penalty for a departure from academic integrity and may also face adverse legal consequences for infringement of intellectual property rights and, with respect to recordings, potentially privacy violations of other students.

Academic Integrity

As an engineering student, you have made a decision to join us in the profession of engineering, a long-respected profession with high standards of behaviour. As future engineers, we expect you to behave with integrity at all times. Please note that Engineers have a duty to:

•Act at all times with devotion to the high ideals of personal honour and professional integrity.

•Give proper credit for engineering work

The standard of behaviour expected of professional engineers is explained in the Professional Engineers Ontario Code of Ethics. Information on policies concerning academic integrity is available in the Queen's University Code of Conduct, in the Senate Academic Integrity Policy Statement, on the Smith Engineering website, and from your instructor.

Departures from academic integrity include plagiarism, use of unauthorized materials or services, facilitation, forgery, falsification, unauthorized use of intellectual property, and collaboration, and are antithetical to the development of an academic community at Queen's. Given the seriousness of these matters, actions which contravene the regulation on academic integrity carry sanctions that can range from a warning or the loss of grades on an assignment to the failure of a course to a requirement to withdraw from the University.

In the case of online or remotely proctored exams, impersonating another student, copying from another student, making information available to another student about the exam questions or possible answers, posting materials to online services, communicating with another person during an exam or about an exam during the exam window, or accessing unauthorized materials, including internet sources and using unauthorized materials, including smart devices, are actions in contravention of academic integrity.

PLAGIARISM/USE OF GENERATIVE ARTIFICIAL INTELLIGENCE (GENAI) TOOLS:

Students must submit their own work and cite the work that is not theirs. Any other use constitutes a Departure from Academic Integrity.

GenAI tools are NOT permitted in this course: Use of GenAI tools are not allowed in any part of student work for this course. Submitting AI-generated content constitutes a Departure from Academic Integrity as defined by University Academic Integrity procedures.

INVALID EXAMS

An exam may be declared invalid in case of an interruption in an in-person examination; if the instructions in a remote or online exam were not followed; if the student uploads wrong materials; or if a situation arises where the integrity of the exam cannot be verified. If an exam is declared invalid, the student may be granted a re-write.

ACADEMIC AND STUDENT SUPPORT

Queen's has a robust set of supports available to you, including the Library, Student Academic Success Services (Learning Strategies and Writing Centre), and Career Services.

Learners are encouraged to visit the Smith Engineering Current Students web portal for information about various other policies such as academic advisors, registration, student exchanges, awards and scholarships, etc. Students are also encouraged to review the information that is available in the EngQ Hub, posted in onQ.

Absences - Academic Considerations

Academic Consideration is a process by which students request validated time away from their academics when they are experiencing an extenuating circumstance or personal circumstances beyond a student's control that interfere with their ability to complete academic requirements.

For additional information and to apply for academic consideration, please review the information on the Smith Engineering's Academic Consideration (Absences) website.

Absences – Requests for Excused Absences for Significant Events (REASE)

Students participating in a Varsity Athletics event, the student reserve forces, a non-varsity event at the provincial, national, or international level, or an event to which you were invited as a distinguished guest can submit a Request for Excused Absence for Significant Event following the REASE process outlined on the Student Affairs webpage.

Students participating in Design Competitions supported by Smith Engineering can submit a Request for Excused Absence for Significant Events (REASE) following the process outlined on the Smith Engineering's Academic Consideration (Absences) website.

All Requests for Excused Absences for Significant Events must be submitted to the Faculty a minimum of 2 weeks prior to the event start date. For additional information and to submit a REASE application, please review the information on the Smith Engineering's Academic Consideration (Absences) website.

Academic Accommodations

Academic Accommodations are available when students need an adjustment to their learning environment due to a reason related to a Human Rights protected ground. Below there is information on the most common Academic Accommodations requested by students. For more information, please see Smith Engineering's Academic Accommodation website.

Religious Accommodations

Students in need of accommodation for religious observance are asked to complete the Religious Observance and Academics Activities form available on Smith Engineering's Academic Accommodation webpage within a week of receiving their course syllabus (each term).

- Alternative assignments are considered a "reasonable accommodation" under the Ontario Human Rights Code
- Please refer to the Queen's Multi-faith Calendar for all of this year's faith dates. Requests for Religious Accommodation will be verified based on the dates in this calendar. If you are requesting dates not listed in this calendar, you may be asked for additional documentation
- Students with questions about their rights and responsibilities regarding religious accommodations can contact chaplain@queensu.ca or Smith Engineering Academic Accommodation and Consideration Advisor at engineering.aac@queensu.ca

Further information can be found on the Smith Engineering's Academic Accommodation webpage

Other Human-Rights Based Accommodations

Engineering students who need accommodation on Human Rights grounds are asked to contact, in confidence, the Smith Engineering Academic Accommodation and Consideration Advisor at mailto:engineering.aac@queensu.ca or to call 613-533-6000 x 78139.

Accommodations for Disabilities

Queen's University is committed to working with students with disabilities to remove barriers to their academic goals. Queen's Student Accessibility Services (QSAS), students with disabilities, instructors, and faculty staff work together to provide and implement academic accommodations designed to allow students with disabilities equitable access to all course material (including in-class as well as exams). If you are a student currently experiencing barriers to your academics due to disability related reasons, and you would like to understand whether academic accommodations could support the removal of those barriers, please visit the QSAS website to learn more about academic accommodation.

To start the registration process with QSAS, click the "**Access Ventus"** button found on the Ventus student portal.

Ventus is an online portal that connects students, instructors, Queen's Student Accessibility Services, the Exam's Office, and other support services in the process to request, assess, and implement academic accommodations. To learn more about Ventus, visit A Visual Guide to Ventus for Students:
https://www.queensu.ca/ventus-support/students/visual-guide-ventus-students

For questions or assistance with requesting Academic Consideration or Accommodation, contact the Smith Engineering Academic Accommodation and Consideration Advisor at engineering.aac@queensu.ca

Every effort has been made to provide course materials that are accessible. For further information on accessibility compliance of the educational technologies used in this course, please consult the links below.

Student Wellbeing

Success in Engineering includes taking care of your wellbeing. If you're experiencing stress, anxiety, burnout, personal challenges, or you're simply not sure where to turn, EngWell is here to help.

EngWell provides confidential support through our Wellness Advisor, Embedded Counsellors, mental health programming, and referrals to academic and campus resources. Whether you need short term guidance, help navigating university supports, or someone to talk to, we encourage you to reach out early.

Educational Technology

Educational Technology	Accessibility Compliance Information
onQ (Brightspace Learning Management System by D2L)	https://www.d2l.com/accessibility/standards/
MS-Teams	https://support.microsoft.com/en-us/office/accessibility-support-for-microsoft-teams-d12ee53f-d15f-445e-be8d-f0ba2c5ee68f
Zoom	https://zoom.us/accessibility

If you find any element of this course difficult to access, please discuss with your instructor how you can obtain an accommodation.

Technical Support

Some basic comfort levels with basic hardware and software skills are required for this course. If you require technical assistance, please contact Technical Support.